KRINI CRUDE • 1-COOL

INDUSTRIAL CHEMICALS • LUBRICANTS DIVISION

1-COOL AL 5000

1-COOL AL 5000 is a semi-synthetic, environment friendly cutting fluid developed for use in light to heavy duty machining of ferrous and non-ferrous metal alloys. It is also suitable for general purpose grinding. Formulated with special lubricity and E.P. additives and rust inhibitors, it delivers extended sump life, reduced downtime, and improved tool longevity.

KEY FEATURES & BENEFITS

- Excellent cooling and lubricating properties for precision surface finish
- Pleasant odour of the emulsions, making it highly user friendly
- High resistance to bacterial attack, reducing odour and maximizing coolant life
- Forms a stable emulsion even in hard water conditions
- Offers excellent rust protection of ferrous components

APPLICATIONS & USES

- Light to heavy duty machining operations
- General purpose metal grinding
- Cast Iron
- Ferrous and Non-ferrous Alloys
- Stainless Steels

TYPICAL TECHNICAL SPECIFICATIONS

Test Parameter	Typical Value / Result	Test Method / Note
Appearance (Concentrate)	Clear Amber fluid	Visual inspection
Emulsion Appearance	Translucent to off white	Diluted in water
Density @ 29.5 °C	0.970 g/ml	Standard density
Thermal Stability Test	Passes	Stable thermal profile
Cast Iron Corrosion Test	No rust	@ 3%, 200 ppm (IP 287)
pH (10% dilution)	9.6	In distilled water

APPLICATION & HANDLING GUIDELINES

- While mixing, always add oil to water and never the reverse, agitating thoroughly to form a stable translucent emulsion.
- Use good quality water. Deionized (D.I.) water gives extended sump life, as high hardness and chlorides reduce product life and corrosion inhibition.
- Ensure machine sumps are thoroughly cleaned and sterilized before charging with fresh coolant mixture.
- Check emulsion concentration regularly using a refractometer (multiplication factor: 1.4) and replenish with premixed emulsion rather than pure water.

Packaging: Standard packing: 26 L pails & 210 L industrial steel drums • **Storage:** Store indoors in original tightly closed containers between 5 °C and 40 °C. Protect from freezing and direct exposure to intense heat or rain. • **Shelf Life:** 12 months from date of manufacture when stored in unopened containers

Technical values are typical and represent current production data. Suitability should be confirmed by site trials. For industrial use only.